How Many Comparisons Does It Take to Flip a Leaderboard? Scale-Free Sensitivity of Bradley–Terry Rankings Across Evaluation Designs

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Abstract

Leaderboards built from pairwise comparisons (crowd votes, expert or model judges, sports results) are fitted with Bradley–Terry (BT) models and read as rankings. We ask how many comparisons must be *changed*—by noise or by an adversary—to flip the BT order of two contestants, and whether the answer depends on the evaluation design. We define the budget of a pair as the number of comparisons that a greedy adversary with exact refitting must remove, relabel or insert, and normalise it by the number of comparisons involving the two contestants. On four designs (Chatbot Arena crowd votes, MT-Bench human experts, MT-Bench GPT-4 judge, 16 NBA seasons) we find: (i) in dense leaderboards most adjacent orders are not statistically resolved and flip with at most 5 edits (62% of adjacent Arena pairs, 98% of NBA pairs), whereas the well-separated MT-Bench pairs need 2–6% of their local comparisons; (ii) the *fraction* of local comparisons needed is proportional to the BT score gap, $f \approx \kappa\,\Delta\theta$ with $\kappa \approx 0.06$ for relabelling (0.08 for NBA) and ≈ 0.12 for removal, nearly constant across designs and across a 32-fold change of the Arena sample size; equivalently, the edit *count* is about $0.2\,z\sqrt{n_{\mathrm{loc}}}$ for z-score z; (iii) a crowd-vs-expert control on six matched models at matched size shows no detectable design effect beyond gap and size; (iv) random removals of the same size never flipped a pair that needs ≥ 5 targeted edits. Designs thus differ mainly in how densely contestants are packed relative to noise, not in intrinsic fragility per unit of score. We state the conditions and limits of these conclusions.

1 Introduction

Model leaderboards that aggregate many "A versus B, which is better?" judgments are used for model selection, release announcements and research comparisons. Their rankings come from a Bradley–Terry (BT) fit [Bradley and Terry, 1952] to the comparison data, often with bootstrap confidence intervals, as on Chatbot Arena [Chiang et al., 2024]. What is much less clear is how strongly a published rank depends on the underlying comparisons, whether the change comes from honest noise, from a few corrupted records, or from deliberate manipulation [Min et al., 2025, Singh et al., 2025], and whether this dependence differs between evaluation designs: anonymous crowd voting, expert annotation, an LLM judge [Zheng et al., 2023], or sports results.

This paper makes the question operational and answers it empirically.

- **Definition.** The *edit budget* $k(i, j)$ of an adjacent pair is the smallest number of unit edits (remove, relabel, or insert a comparison) found by a greedy adversary, with exact BT refitting, that reverses the estimated order of i and j. To compare boards of different size we normalise by n_{loc}, the number of comparisons involving i or j (Sec. 3).

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- **Evidence.** Four designs from public data: Chatbot Arena crowd votes (64 models, 57,477 votes), MT-Bench human-expert and GPT-4-judge pairs (6 models), and 16 NBA regular seasons (about 30 teams, 1,230 games each). We also run a matched control (Arena votes restricted to the six MT-Bench models), subsampling, random-edit, and robustness experiments.
- **Findings.** (a) Sensitivity is not one number per leaderboard: most adjacent pairs in dense boards are statistical ties and need ≤ 5 edits, while resolved pairs need a few percent of their local data. (b) The fraction of local comparisons needed is proportional to the BT gap with a design-independent constant to within a factor of about 1.4, and is independent of the amount of data. (c) Hence statistical resolution (a z-score) and robustness to a fractional adversary are different things: more data resolves more pairs but does not make a given gap harder to overturn in relative terms.

Our conclusions are conditional: they concern first-order-adjacent ranks in BT models fitted to the specific snapshots listed in Sec. 4; the adversary is a heuristic upper bound on the true minimum; and we do not model how a real attacker could realise the edits.

2 Related work

Paired-comparison leaderboards. The BT model [Bradley and Terry, 1952] underlies Chatbot Arena [Chiang et al., 2024], which collects pairwise human votes and reports ranks with bootstrap intervals. MT-Bench [Zheng et al., 2023] provides expert human and GPT-4 pairwise judgments on 80 questions for six public models; we use both. **Manipulation and distortion.** Min et al. [2025] study how voting strategies can raise or lower a model's Arena rank, and Singh et al. [2025] document evaluation practices (private testing, unequal sampling) that distort the leaderboard. Our analysis is complementary: instead of a particular attack or practice we ask, for a fixed snapshot, how large *any* edit set must be, and compare designs. **Data-removal robustness.** Influence functions [Koh and Liang, 2017] approximate how a model changes when training points are removed, and Broderick et al. [2020] use such approximations to find the small data subset whose removal changes an applied conclusion ("Approximate Maximum Influence Perturbation"). We use the same linearisation to propose edits but refit exactly after each chunk, extend the edit set to relabelling and insertion, and apply it across evaluation designs. We did not consult every listed paper in full (App. A); in particular we did not read the leaderboard-specific work on dropping preferences that appears in the task's reading list, so we make no priority claim relative to it.

3 Method

Model. A comparison s between a_s and b_s has outcome $y_s \in \{1, \frac{1}{2}, 0\}$ (win for a_s, tie, win for b_s). Ties are counted as half a win each, as in Elo/BT leaderboards. We fit $\hat{\theta} = \arg\min_\theta -\sum_s [y_s \log \sigma(\theta_{a_s} - \theta_{b_s}) + (1 - y_s) \log \sigma(\theta_{b_s} - \theta_{a_s})] + \frac{\lambda}{2}\|\theta\|^2$ with $\lambda = 10^{-3}$ (fixing the additive gauge; results are unchanged for $\lambda \in [10^{-5}, 1]$, App. B) by Newton's method on pair-aggregated counts. Contestants are ranked by $\hat{\theta}$; the Hessian is H.

Pair sensitivity. For ranks r and $r+d$ ($d = 1$ unless stated) with contestants i, j, gap $\Delta\hat{\theta} = {}_i - {}_j > 0$. A *unit edit* is: *remove* one comparison; *relabel* one comparison to another outcome; or *insert* one new comparison of any pair and outcome (rigging-style; not meaningful for fixed schedules such as the NBA, where we report it only for completeness). The budget $k_{\mathrm{mode}}(i, j)$ is the smallest number of edits found that makes ${}_i < {}_j$ after refitting.

Greedy adversary. Let $c = e_i - e_j$ and $v = H^{-1}c$. To first order, an edit of a comparison with design vector $x = e_a - e_b$ and fitted win probability p changes the contrast $c^\top\theta$ by $v^\top x\,(p - y)$ (removal), $v^\top x\,(y' - y)$ (relabel $y \to y'$) or $v^\top x\,(y' - p)$ (insertion). Each round ranks all (pair, outcome) cells by predicted decrease, applies a chunk of $\max(1, \lfloor \frac{1}{2}\,\text{margin}/\text{best gain} \rfloor)$ edits subject to cell counts, refits exactly, and stops when the order is reversed; chunks shrink to one edit near the end. Edits may touch any comparison, not only those of i and j; n_{loc} is just a normaliser (for insertions it is the original count). The returned k is therefore an *upper bound* on the true minimum. On 150 (removal) and 152 (relabel) random tiny instances (3 contestants, 20–39 votes) where exhaustive search was

Table 1: Boards studied. "bootstrap τ": mean Kendall τ between the full-data ranking and bootstrap rankings; "top-1 kept": fraction of bootstrap fits with the same top contestant; last column: share of adjacent pairs with $z \geq 2$ (bootstrap 95% interval). NBA entries: median over 16 seasons (range).

Design	#boards	players	votes N	ties	bootstrap τ	top-1 kept	adjacent pairs $z \geq 2$
Arena (crowd votes)	1	64	57477	31%	0.92	0.47	6% [2, 13]
MT-Bench (human)	1	6	3355	23%	0.96	1.00	80% [40, 100]
MT-Bench (GPT-4 judge)	1	6	2400	25%	1.00	1.00	100% [100, 100]
NBA (games)	16	30	1230	0%	0.81 (0.76–0.83)	0.50 (0.28–0.89)	0% [0, 1]

feasible, greedy matched the exact minimum in 96.7% and 99.3% of cases and never beat it; its worst ratio was 2.0 (App. B).

Normalisation across designs. Raw counts depend on board size. We report (i) $f = k/n_{\mathrm{loc}}$, where $n_{\mathrm{loc}} = n_i + n_j - n_{ij}$ is the number of comparisons involving i or j; (ii) the BT gap $\Delta\hat{\theta}$ in logits; and (iii) the z-score $z = \Delta\hat{\theta}/\hat{\sigma}_{ij}$, with $\hat{\sigma}_{ij}$ the standard deviation of $_i - _j$ over B=200 nonparametric bootstrap resamples of the votes. Two empirical constants summarise a design: $\kappa = f/\Delta\hat{\theta}$ and $c = k/(z\sqrt{n_{\mathrm{loc}}})$.

Proposition 1 (First-order scale of the budget)**.** *Let $\sigma_c^2 = c^\top H^{-1} c$ and $s_x^2 = x^\top H^{-1} x$. If a set of edits $\{(x_s, \delta_s)\}$ reverses the pair in the linearisation, then $\sum_s |\delta_s|\, s_{x_s} \geq \Delta\hat{\theta}/\sigma_c$.*

Proof sketch. The linearised change of $c^\top\theta$ is $\sum_s \delta_s\, c^\top H^{-1} x_s$; it must be at most $-\Delta\hat{\theta}$. Since $H^{-1} \succeq 0$, Cauchy–Schwarz gives $|c^\top H^{-1} x_s| \leq \sigma_c s_{x_s}$. □

The proposition is only heuristic (the true map is nonlinear and H^{-1} changes as edits are made) but explains the form of our empirical findings: a contrast with standard error $\sigma_c \propto 1/\sqrt{n_{\mathrm{loc}}}$ separated by z standard errors needs an "edit mass" of at least z in units of per-comparison standard errors, i.e. a count that grows like $z\sqrt{n_{\mathrm{loc}}}$, and a *fraction* k/n_{loc} that grows like $\Delta\hat{\theta}$ once $\sigma_c\sqrt{n_{\mathrm{loc}}}$ is roughly constant.

4 Data and protocol

Table 1 lists the boards (sources, versions and checksums in App. A). **Arena (crowd):** the 55k human-preference snapshot of Chatbot Arena, 64 models, 31% ties (including "both bad"). **MT-Bench (human):** 3,355 judgments by experts and authors on 80 two-turn questions; **MT-Bench (GPT-4 judge):** 2,400 GPT-4 pairwise verdicts on the same questions ("inconsistent" verdicts counted as ties). **NBA (sports):** one board per regular season 2000–2015 (franchise level, no home-court term), chosen before looking at results as the 29–30-team era. The four designs differ in who judges, number of contestants (6 to 64), votes per contestant, tie rates, and how dense the ability scale is. Throughout, the MT-Bench boards have 6 contestants, so "adjacent pairs" there number only 5 each; their statistics are correspondingly coarse.

Uncertainty. Intervals for fractions and medians are percentile bootstrap intervals over pairs (over seasons' pairs for NBA, resampled by season). Bootstrap standard deviations $\hat{\sigma}_{ij}$ use $B = 200$ vote-level resamples (question-clustered resampling is reported in App. B). All randomness is seeded; the whole pipeline runs in a few minutes on CPU. **Exploratory vs. confirmatory.** The study design (boards, adversaries, normalisation by n_{loc}, z) was fixed before running. The proportionality $f \propto \Delta\hat{\theta}$ and the constants κ were identified post hoc from the main pair table; later subsampling and matched-control runs were analysed for κ afterwards, so only the agreement across designs (not the subsample series, which reuses the same Arena votes) counts as a consistency check. Thresholds ($z \geq 2$, $k \geq 5$, $\Delta\hat{\theta} > 0.05$, $\epsilon \in \{0.5, 1, 2, 5\}\%$) were set by us without pre-registration; $k \geq 5$ and $\Delta\hat{\theta} > 0.05$ merely exclude pairs where integer rounding or near-zero gaps dominate. We report descriptive intervals, not hypothesis tests, so no multiple-testing correction is applied.

Table 2: Share (%) of adjacent pairs that a relabelling adversary can flip using at most a fraction ϵ of the votes involving the two contestants. 95% bootstrap intervals; tiny n (MT-Bench: 5 pairs) makes the intervals wide.

Design	pairs	ϵ=0.5%	ϵ=1%	ϵ=2%	ϵ=5%
Arena (crowd votes)	all adjacent (n=63)	84 [75,92]	97 [92,100]	100 [100,100]	100 [100,100]
MT-Bench (human)	all adjacent (n=5)	20 [0,60]	20 [0,60]	20 [0,60]	80 [40,100]
MT-Bench (GPT-4 judge)	all adjacent (n=5)	0 [0,0]	0 [0,0]	0 [0,0]	80 [40,100]
NBA (games)	all adjacent (n=453)	0 [0,0]	68 [65,71]	94 [92,96]	100 [99,100]
Arena (crowd votes)	$z \geq 2$ (n=4)	0 [0,0]	50 [0,100]	100 [100,100]	100 [100,100]
MT-Bench (human)	$z \geq 2$ (n=4)	0 [0,0]	0 [0,0]	0 [0,0]	75 [25,100]
MT-Bench (GPT-4 judge)	$z \geq 2$ (n=5)	0 [0,0]	0 [0,0]	0 [0,0]	80 [40,100]
NBA (games)	$z \geq 2$ (n=1)	0 [0,0]	0 [0,0]	0 [0,0]	0 [0,0]

Table 3: The top-ranked contestant vs. rank 2: gap, bootstrap z, local votes, and edits needed to reverse (relabel / remove). NBA: median over seasons (range).

Board	gap $\hat{\theta}_1 - \hat{\theta}_2$	z	n_{loc}	k_{relabel}	k_{remove}	$k_{\mathrm{relabel}}/n_{\mathrm{loc}}$
Arena (crowd votes)	0.00	0.1	8413	1	2	0.01%
MT-Bench (human)	0.38	4.8	1825	46	92	2.52%
MT-Bench (GPT-4 judge)	0.82	8.7	1440	61	143	4.24%
NBA (games)	0.15 (0.00–0.62)	0.4 (0.0–1.5)	160 (131–162)	2 (0–8)	3 (0–9)	1.39 (0.00–4.94)%

5 Results

5.1 Most adjacent orders in dense boards are inside the noise—and cost a handful of edits

Table 1 shows how different the boards are statistically: 80–100% of the adjacent MT-Bench pairs are resolved ($z \geq 2$), but only 6% (4 of 63) in Arena and 0.2% (1 of 459) in NBA. The Arena leader is not resolved either: `gpt-4-1106-preview` and `gpt-4-0125-preview` differ by 0.004 logits, and the top-1 identity survives only 47% of bootstrap fits (NBA: 50%, range 28–89%).

Table 2 reports the share of adjacent pairs flippable by relabelling at most a fraction ϵ of their local votes. At $\epsilon = 1\%$ this is 97% of Arena pairs and 68% of NBA pairs, but 20% and 0% for the two MT-Bench boards; by $\epsilon = 5\%$ nearly everything flips (MT-Bench: 4 of 5 pairs). In absolute terms $k_{\mathrm{relabel}} \leq 5$ for 62% of Arena and 98% of NBA adjacent pairs and for none of the MT-Bench pairs. Restricting to resolved pairs ($z \geq 2$, lower half of Table 2) changes the picture: the 4 resolved Arena pairs need 0.5–2% of local votes (e.g. `gpt-4-0125-preview` vs `gpt-4-0314`: 105 relabels among 5,265 local votes, $z = 7.7$) and resolved MT-Bench pairs need 2.5–6%. The single resolved NBA pair (Wizards, 2012) needs about 8%. Table 3 gives the leader: in MT-Bench, dethroning GPT-4 needs 46 (human) or 61 (GPT-4 judge) relabels, 2.5% and 4.2% of its local comparisons; the BT-top team of the median NBA season is overtaken by 2 relabels (range 0–8), and in 12 of 16 seasons has $z < 1$ over the runner-up.

5.2 A scale-free regularity: the fraction of votes to edit is proportional to the gap

Figure 1(a) plots $f = k_{\mathrm{relabel}}/n_{\mathrm{loc}}$ against the gap for every adjacent pair of every board. Above the integer floor $f \geq 1/n_{\mathrm{loc}}$ (visible for NBA and near-ties), the points follow a line through the origin in log–log coordinates: for pairs with $\Delta\hat{\theta} > 0.05$ and $k \geq 5$ (42 pairs), a log–log regression has slope 1.06 (95% cluster-bootstrap interval $[0.43, 1.15]$) for relabelling, 1.01 $[0.68, 1.05]$ for removal and 1.03 $[0.73, 1.07]$ for insertion, with residual sd 0.29 in $\log f$. The proportionality constants, Table 4, are $\kappa_{\mathrm{relabel}} \approx 0.06$ (Arena 0.059, MT-Bench human 0.067, GPT-4 judge 0.056, NBA 0.082), $\kappa_{\mathrm{remove}} \approx 0.11$–$0.14$ and $\kappa_{\mathrm{insert}} \approx 0.10$–$0.13$. Interpreted: to reverse a gap of 0.3 logits (about a 7-point win-probability difference near $\frac{1}{2}$), an adversary must relabel about 1.8% of the votes involving the two contestants, removal needs about twice as many edits (median ratio relabel/remove 0.60, 10–90% range 0.46–0.71) and insertion about 0.9 of the removal cost. Design differences in κ are at most a factor of 1.5 (NBA vs. MT-Bench GPT-4), and the NBA value is the only one clearly outside the others' intervals; we do not have a tested explanation for it (a test of the hypothesis that ties explain the difference, by dropping ties, did not support it, App. B).

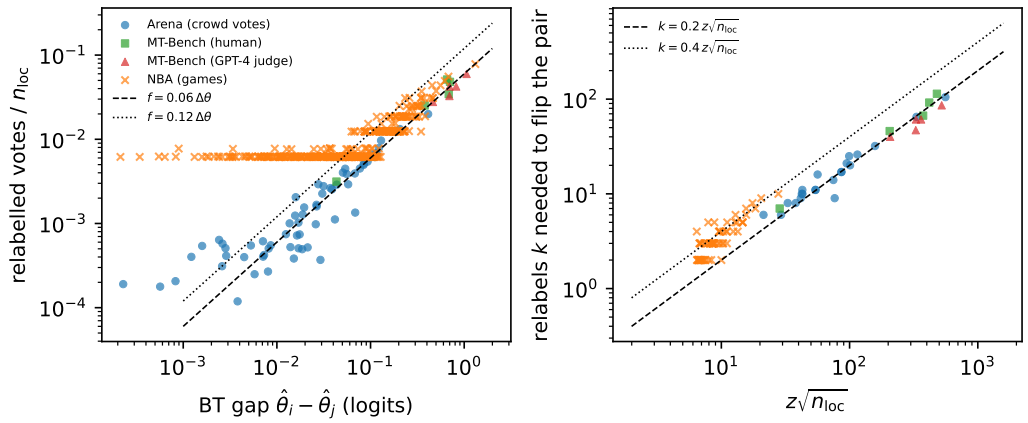


Figure 1: (a) Fraction of local votes relabelled to flip an adjacent pair vs. BT gap, all pairs of all boards (453 NBA, 63 Arena and 5+5 MT-Bench pairs; 6 NBA pairs with identical fitted scores are omitted). The horizontal run at $f \approx 6 \times 10^{-3}$ is the one-edit floor for NBA. (b) Number of relabels vs. $z\sqrt{n_{\mathrm{loc}}}$ for pairs with $z \geq 0.5$.

Table 4: $\kappa = f/\Delta\hat{\theta}$, the fraction of the contestants' votes to edit per logit of gap, median over pairs with $\Delta\hat{\theta} > 0.05$ and $k_{\mathrm{relabel}} \geq 5$ (95% bootstrap interval).

Design	pairs	$\kappa_{\mathrm{relabel}}$	κ_{remove}	κ_{insert}
Arena (crowd votes)	18	0.059 [0.058, 0.064]	0.116 [0.111, 0.134]	0.105 [0.093, 0.116]
MT-Bench (human)	4	0.067 [0.050, 0.078]	0.142 [0.106, 0.157]	0.134 [0.100, 0.156]
MT-Bench (GPT-4 judge)	5	0.056 [0.047, 0.061]	0.123 [0.095, 0.150]	0.113 [0.094, 0.122]
NBA (games)	15	0.082 [0.074, 0.096]	0.112 [0.101, 0.124]	0.101 [0.097, 0.117]

Link to statistical resolution. Combining $f = \kappa\Delta\hat{\theta}$ with $\Delta\hat{\theta} = z\hat{\sigma}_{ij}$ predicts $k = c\,z\sqrt{n_{\mathrm{loc}}}$ if $\hat{\sigma}_{ij}\sqrt{n_{\mathrm{loc}}}$ is roughly constant. It is: its median is 3.7 (Arena), 3.4 (MT-Bench human), 3.1 (GPT-4 judge), 4.5 (NBA). Figure 1(b) confirms $k \approx c\,z\sqrt{n_{\mathrm{loc}}}$ over almost three decades with c_{relabel} (median, $z \geq 1$) equal to 0.20, 0.22, 0.17 and 0.38 for the four designs and c_{remove} of 0.40, 0.46, 0.42 and 0.54 (intervals in App. B). A pooled regression of $\log k_{\mathrm{relabel}}$ on $\log z$ and $\log n_{\mathrm{loc}}$ (36 pairs with $z \geq 1$) gives exponents 0.89 on z ($[0.75, 1.29]$) and 0.32 on n_{loc} ($[0.15, 1.80]$); for removal 0.97 ($[0.78, 1.05]$) and 0.42 ($[-0.04, 0.49]$). The $\sqrt{n_{\mathrm{loc}}}$ exponent is thus imprecisely determined (the 36 pairs vary mostly in z), and the exponent 0.5 is consistent with removal but at the edge for relabelling.

5.3 Rank displacement and random versus targeted edits

Moving a contestant d places down costs more roughly linearly in d (Figure 2, left): the median relabel fraction for $d = 1$ vs. $d = 5$ is 0.10% vs. 0.75% (Arena), 3.4% vs. 22% (MT-Bench human), 4.2% vs. 30% (GPT-4 judge) and 0.63% vs. 3.7% (NBA). Pushing GPT-4 from first to last place in MT-Bench takes 423 (human) and 430 (GPT-4 judge) relabels, i.e. 22% and 30% of its local votes against the last-ranked model.

Random edits are far less damaging (Fig. 2, right). Relabelling 1% of all votes at random leaves Kendall τ at 0.984 (Arena), 1.00 (MT-Bench human/GPT-4) and 0.96 (NBA), while the sampling noise of the boards, as measured by bootstrap τ, equals what random relabelling of roughly 8–20% of the votes would produce (bootstrap τ Arena 0.92, NBA 0.80, MT-Bench human 0.96; the GPT-4-judge board has bootstrap $\tau = 1.00$). Random removal of exactly the adversary's budget k_{remove} votes (50 repetitions per pair) flipped none of the 92 pairs with $k_{\mathrm{remove}} \geq 5$ in any repetition; it flips only some of the pairs with near-zero gaps that need 1–4 edits. The share of adjacent pairs flipped by random 1% relabelling is 15% (Arena) and 19% (NBA), i.e. these flips are essentially the pairs that are already coin flips under resampling (mean bootstrap flip probability 0.41 for $z < 1$, 0.11 for $1 \leq z < 2$, 0.003 for $z \geq 2$). Targeted edits thus do not create instability in dense boards; they

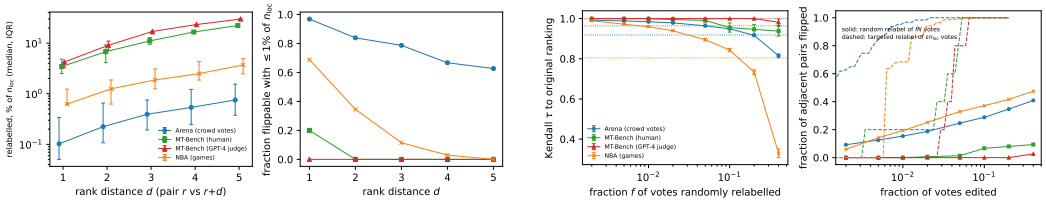


Figure 2: Left: cost of displacing a contestant by d ranks (median and interquartile range over pairs; right sub-panel: share flippable with $\leq 1\%$ of local votes). Right: random relabelling of a fraction f of all votes (mean Kendall τ to the original ranking, 95% intervals over 30 repetitions, 5 per NBA season; dotted: bootstrap τ) and fraction of adjacent pairs flipped, random (solid) vs. targeted (dashed, per-pair adversary).

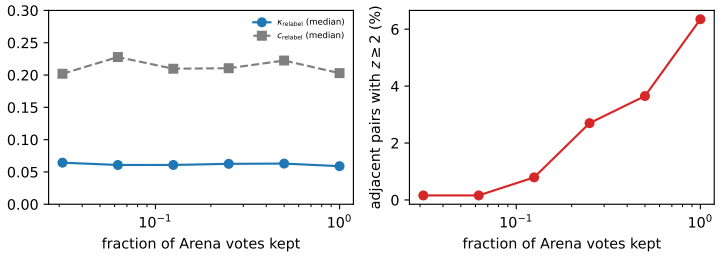


Figure 3: Arena votes subsampled. Left: median κ_{relabel} and c_{relabel}. Right: share of adjacent pairs with $z \geq 2$.

exploit what resampling already leaves undecided and then, for resolved pairs, a few percent of local votes.

5.4 Does more data help? Subsampling and a matched crowd-vs-expert control

Subsampling. We keep a random $1/32, \ldots, 1/2$ of the Arena votes (10 repetitions each) and recompute everything. κ_{relabel} stays at 0.061–0.064 for kept fractions $1/32$ to $1/2$ (3 to 115 qualifying pairs per setting) and is 0.059 at full size, a 32-fold range (Figure 3, left; the subsamples reuse the same votes, so this tests dependence on size, not independent replication), and c_{relabel} at 0.20–0.23, while the share of resolved adjacent pairs rises from 0.2% to 6.3%. More data therefore resolves more pairs but does not change the *fraction* of votes needed per logit of gap; the absolute count grows in proportion to n_{loc}.

Matched control. To separate evaluation design from board size and contestant set, we restrict Arena votes to six models that match the MT-Bench set (`gpt-4-0314`, `claude-1`, `gpt-3.5-turbo-0314`, `vicuna-13b`, `alpaca-13b`, `llama-13b`; versions are the nearest available, not identical), leaving 1,367 votes with 27% ties, and subsample MT-Bench to the same size (30 repetitions). The BT scores of the two boards agree closely (e.g. gpt-4 1.09 vs. 1.02, llama-13b -1.53 vs. -1.45 for Arena-6 vs. MT-Bench human), though the crowd board separates claude-v1 from gpt-3.5-turbo more (0.34 vs. 0.04 logits). Table 5 shows the five adjacent pairs: the relabel fraction is 2.4–5.7% (Arena-6, median $\kappa = 0.071$), 2.5–4.9% for MT-Bench human apart from a near-tied pair (median $\kappa = 0.068$) and 2.8–6.0% for the GPT-4 judge ($\kappa = 0.057$). The three median κ values lie within a factor of 1.25 of each other, smaller than the pair-to-pair spread inside the Arena-6 board (0.050–0.101). We conclude that, for these six models, crowd, expert and model-judge designs show no detectable difference in κ or c once gap and size are accounted for; this is one contestant set and one snapshot per design, with 5 pairs per board.

5.5 Validity checks

The edit budgets are insensitive to the ridge $\lambda \in [10^{-5}, 1]$ (median ratio to the baseline 1.0; 5–95% range 0.90–1.10 only at $\lambda = 1$). One-shot linearised estimates (no refitting, as in AMIP-style

Table 5: Matched control: Arena votes restricted to six models (1,367 votes) versus MT-Bench subsampled to 1,367 votes (mean over 30 subsamples; pair identity by rank).

Board	adjacent pair	gap	z	n_{loc}	k_{relabel}	k/n_{loc} (%)
Arena-6 (crowd, N=1367)	gpt-4 / claude-v1	0.30	3.5	882	27	3.1
Arena-6 (crowd, N=1367)	claude-v1 / gpt-3.5-turbo	0.34	3.3	872	21	2.4
Arena-6 (crowd, N=1367)	gpt-3.5-turbo / vicuna-13b-v1.2	0.46	4.8	858	28	3.3
Arena-6 (crowd, N=1367)	vicuna-13b-v1.2 / alpaca-13b	0.76	7.5	865	49	5.7
Arena-6 (crowd, N=1367)	alpaca-13b / llama-13b	0.75	4.8	607	23	3.8
MT-Bench human, subsampled to 1367	rank 1–2	0.37	3.2	800	20	2.5
MT-Bench human, subsampled to 1367	rank 2–3	0.07	0.6	908	4	0.5
MT-Bench human, subsampled to 1367	rank 3–4	0.64	6.2	872	42	4.8
MT-Bench human, subsampled to 1367	rank 4–5	0.71	6.5	776	38	4.9
MT-Bench human, subsampled to 1367	rank 5–6	0.69	5.4	793	28	3.6
MT-Bench GPT-4, subsampled to 1367	rank 1–2	0.79	6.4	817	34	4.1
MT-Bench GPT-4, subsampled to 1367	rank 2–3	0.46	4.4	818	23	2.8
MT-Bench GPT-4, subsampled to 1367	rank 3–4	0.70	6.6	823	35	4.3
MT-Bench GPT-4, subsampled to 1367	rank 4–5	1.05	9.3	823	49	6.0
MT-Bench GPT-4, subsampled to 1367	rank 5–6	0.70	5.9	823	27	3.3

approximations) are close to the refit greedy result: median ratio one-shot/greedy 1.06 (removal; 10–90% 1.0–1.33), 1.0 (relabel) and 0.91 (insertion; 92 pairs with $k_{\mathrm{remove}} \geq 5$). Dropping ties gives $c_{\mathrm{relabel}} = 0.21$–$0.24$, similar to the baseline. Question-clustered resampling of MT-Bench (judgments of the same question are not independent) shrinks z by a factor of 0.62 (human) and 0.57 (GPT-4 judge), which raises c_{relabel} to 0.35 and 0.31 but leaves κ (which does not use z) unchanged; the resolved/unresolved split for MT-Bench largely remains ($z \geq 3$ for four of five human pairs and all five GPT-4-judge pairs; the rank 2–3 human pair stays unresolved with $z = 0.35$). Details are in App. B.

6 Discussion and limitations

What the evidence supports. For BT leaderboards on these four designs, the edits needed to reverse an adjacent order are (a) trivially few (≤ 5) whenever the pair is not statistically resolved, which is most pairs in dense boards, and (b) about $0.06\,\Delta\hat{\theta}\,n_{\mathrm{loc}}$ relabels, $0.12\,\Delta\hat{\theta}\,n_{\mathrm{loc}}$ removals, for resolved pairs, with a design-dependent constant within a factor of 1.5. "Sensitivity" is therefore best read as a property of the gap and of local data volume; the evaluation design enters mainly through how many contestants sit within a few $\hat{\sigma}$ of each other. A practical consequence is that a single global robustness number for a leaderboard is uninformative; a fraction-of-local-votes budget as a function of the gap (the constant κ) transfers across boards.

Limitations. (1) *Greedy upper bounds.* k is an upper bound on the minimum; exhaustive checks cover only 3-contestant toy instances, so slack on real boards is not excluded (although one-shot and refit estimates agree). (2) *Realisability.* Relabelling and removal are not actions an Arena voter can perform; insertion is closer to rigging, but we do not model voter-side constraints such as sampling of battles, identification of the target model, or detection, so budgets are lower bounds on attacker effort, not predictions of attainable rank changes. (3) *Few boards.* Only three non-sports boards exist here; the MT-Bench boards have 6 contestants and 5 adjacent pairs each, so per-design intervals for them are wide, and the Arena snapshot is from 2023–24 with models that are no longer current. The NBA board differs in kind (fixed schedule, no ties) and is the only one with a visibly larger κ. The Arena-6 control matches contestants only approximately. (4) *Model assumptions.* We use plain BT with ties as half-wins and no home advantage, prompt effects or position bias; deployed leaderboards add features or other estimators, and conclusions may change. (5) *z-scores* come from vote-level bootstrap and overstate resolution for clustered data (App. B). (6) *Statistics.* The pooled regressions use 36–42 non-independent pairs and cluster-bootstrap intervals over only 19 boards; the $\sqrt{n_{\mathrm{loc}}}$ exponent is imprecise. (7) Only adjacent pairs and displacement of up to five ranks were studied; edits that change set membership of a top-m group or global metrics (e.g. τ) at minimum cost were not optimised. (8) We did not study temporal drift, duplicated prompts or non-BT estimators.

Broader impact. The analysis can inform defenders (where to add votes or report ties) but also quantifies how little data is needed to reorder close contestants; we report only aggregate budgets on public data and release no attack tooling beyond the analysis code.

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A Reproduction notes and data

Data. Arena 55k: `lmarena-ai/arena-human-preference-55k`, commit `18c2983...`, apache-2.0, sha256 of `train.csv` checked (`0692154a...`). MT-Bench: `lmsys/mt_bench_human_judgments`, commit `f7d2896...`, cc-by-4.0 (human file sha256 `4877bc46...` verified). NBA: FiveThirtyEight `nba-elo/nbaallelo.csv`, commit `4c1ff5e...`, CC-BY-4.0, sha256 `d46ed354...` verified; one row per game (`_iscopy=0`), regular season only. Optional datasets in the task (Arena 140k, LLM-judge Arena, WebDev Arena) were not used. Files live in `data/` and are not part of the code release. The released `README.md` lists commands, seeds and the script that produces each table and figure.

LLM assistance. An AI coding/writing assistant (Claude) planned the study, wrote the code and drafted this text; all reported numbers were produced by the released scripts.

Literature. We read the abstracts of Chiang et al. [2024], Zheng et al. [2023], Koh and Liang [2017], Broderick et al. [2020], Min et al. [2025], Singh et al. [2025] through the arXiv API; Bradley and Terry [1952] is cited from the task's reading list without re-reading. One paper in the starter list (on dropping preferences from LLM leaderboard data) was deliberately not retrieved or read, to keep the study independent; no restricted material was encountered.

B Additional results and checks

Greedy vs. exhaustive. 150 random instances for removal (exhaustive search up to 6 removals) and 152 for relabelling (up to 4 relabels) with 3 contestants and 20–39 comparisons, kept when the exact minimum was within the search limit: greedy equals exact in 96.7% / 99.3% of instances, never below, mean ratio 1.022 / 1.007, maximum 2.0. These instances are far smaller than the real boards.
Ridge. $\lambda \in \{10^{-5}, 10^{-3}, 10^{-1}, 1\}$, five boards (Arena, both MT-Bench, NBA 2010 and 2015): median budget ratio 1.0 for all λ; κ_{relabel} 0.0607, 0.0607, 0.0607, 0.0617.
Ties dropped (Arena, MT-Bench): c_{relabel} 0.225 (Arena, 12 pairs), 0.243 (human, 4), 0.207 (GPT-4, 5); κ_{relabel} 0.053, 0.056, 0.033 respectively (the last from 5 pairs on a board that loses 25% of its data).
Question-clustered bootstrap (300 resamples of the 80 questions): z for the five adjacent MT-Bench-human pairs 3.0, 0.35, 6.9, 5.8, 5.3 (vote-level: 4.8, 0.60, 10.2, 9.6, 8.5); GPT-4 judge 5.2, 3.1, 5.4, 7.0, 5.9 (vote-level 8.7, 5.5, 9.5, 13.8, 8.7).

Table 6: $c = k/(z\sqrt{n_{\mathrm{loc}}})$ by adversary: median over pairs with $z \geq 1$ or $z \geq 2$ (95% bootstrap interval). NBA has a single pair with $z \geq 2$.

$z \geq$	Design	pairs	c_{relabel}	c_{remove}	c_{insert}
1	Arena (crowd votes)	14	0.20 [0.20, 0.22]	0.40 [0.39, 0.44]	0.36 [0.33, 0.39]
1	MT-Bench (human)	4	0.22 [0.18, 0.24]	0.46 [0.38, 0.49]	0.44 [0.36, 0.47]
1	MT-Bench (GPT-4 judge)	5	0.17 [0.14, 0.19]	0.42 [0.29, 0.43]	0.34 [0.29, 0.38]
1	NBA (games)	13	0.38 [0.34, 0.45]	0.54 [0.51, 0.60]	0.50 [0.48, 0.52]
2	Arena (crowd votes)	4	0.20 [0.19, 0.23]	0.40 [0.39, 0.50]	0.37 [0.21, 0.41]
2	MT-Bench (human)	4	0.22 [0.18, 0.24]	0.46 [0.38, 0.49]	0.44 [0.36, 0.47]
2	MT-Bench (GPT-4 judge)	5	0.17 [0.14, 0.19]	0.42 [0.29, 0.43]	0.34 [0.29, 0.38]
2	NBA (games)	1	–	–	–

Arena subsampling. Median $\kappa_{\mathrm{relabel}}$ by kept fraction $1/32, 1/16, 1/8, 1/4, 1/2, 1$: 0.064, 0.061, 0.061, 0.063, 0.063, 0.059; qualifying pairs 3, 12, 31, 66, 115, 18 (10 repetitions for fractions < 1).
Adversary comparison. Median relabel/remove budget ratio 0.60 (10–90%: 0.46–0.71); insert/remove 0.90.

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